

**Juan David Muñoz-Bolaños**

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)  
+43 676 9115145  
juan.munoz@i-med.ac.at

**Mühlbauer GmbH & Co. KG**

Frau Tugce Zabun  
Josef-Mühlbauer-Platz 1  
93426 Roding  
Deutschland

**Subject: Bewerbung als Application Engineer Image Processing (m/w/d)**

Sehr geehrte Frau Zabun,

als Physikingenieur mit Fokus auf industrieller Bildverarbeitung bewerbe ich mich für die Position in Roding. Bei INTECOL S.A.S. habe ich bereits Vision-Lösungen (Halcon/Cognex) für Kunden wie Renault realisiert, wobei die präzise Abstimmung einzelner Optik- und Beleuchtungskomponenten entscheidend war.

In meiner Promotion habe ich komplexe Mikroskopsysteme aufgebaut und hochpräzise optomechanische Komponenten (Thorlabs, Edmund Optics) qualifiziert. Diese Hands-on-Erfahrung in der Justage und Hardware-Software-Integration ermöglicht es mir, Vision-Komponenten zielsicher zu optimieren und QA-Prozesse in der Montage zu sichern.

Bestens in die DACH-Arbeitskultur integriert (B2), stehe ich Ihnen gerne auch für internationale Serviceeinsätze zur Verfügung. Ich freue mich darauf, meine Praxiserfahrung und Begeisterung für Optik in Ihr Team einzubringen.

Mit freundlichen Grüßen,

**Juan David Muñoz-Bolaños**

Application Engineer Image Processing  
Medizinische Universität Innsbruck

# Juan David Muñoz-Bolaños

## Application Engineer Image Processing

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## PROFESSIONAL SUMMARY

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Experienced Physics Engineer specialized in industrial machine vision, optical design, and hardware-software integration. Proven background in commissioning custom vision solutions on production lines. Expert in choosing the right illumination, cameras, and lenses to ensure high-performance quality assurance in automated manufacturing environments.

## TECHNICAL COMPETENCIES

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### Machine Vision Hardware

Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

### Image Processing Libraries

Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

### System Alignment & Justage

Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

### Automation & Integration

Skilled in automating optical systems with Python, C++, and LabVIEW. Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

### R&D Methodology

Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

## PROFESSIONAL EXPERIENCE

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### INTECOL S.A.S.

*Junior Machine Vision System Engineer*

Medellín, Colombia

*Mar 2018 – Mar 2019*

- **Vision Pipeline Engineering:** Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.
- **Industrial Integration:** Commissioned Cognex VisionPro systems for auto-part recognition at Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.
- **QA & Feasibility:** Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

### Medical University of Innsbruck

*PhD Researcher – Optical Metrology & Automation*

Innsbruck, Austria

*Jan 2022 – Present*

- **System Build & Justage:** Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.
- **Adaptive Optics Control:** Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.
- **Vendor & Component QM:** Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

### Leibniz Institute of Photonic Technology (IPHT)

*R&D Researcher (Photonics & Data)*

Jena, Germany

*Jun 2020 – Dec 2021*

- **Instrumentation Design:** Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.

- **Data Evaluation Hooks:** Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

## EDUCATION

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**Medical University of Innsbruck**  
*Ph.D. in Adaptive Optics & Microscopy*

Austria  
*Jan 2022 – Present*

**Friedrich Schiller University Jena**  
*M.Sc. in Photonics*

Germany  
*2019 – 2021*

**University of Cauca**  
*B.Sc. in Physics Engineering*

Colombia  
*2012 – 2018*

## PUBLICATIONS & AWARDS

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- **Awards:** ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

## LANGUAGES

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English (Professional),  
German (B2 - Working Proficiency),  
Spanish (Native)

## REFERENCES

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**Prof. Monika Ritsch-Marte**  
*Head of Institute*  
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

**Dr. Christoph Krafft**  
*Spectroscopy Group Leader*  
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

**Juan Munoz Bolanos**

has participated in the

**ZEISS European  
Summer School**

Lithography Optics &  
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

*Thomas Marschner*

Dr. Thomas Marschner  
Program Manager Funding Projects

*Thomas Stämmler*

Dr. Thomas Stämmler  
CTO & Head of Product Strategy





# URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die  
Physikalisch-Astronomische Fakultät  
mit dieser Urkunde

**Herrn Juan David Munoz Bolanos**  
**geb. 04.06.1994 in La Cruz (Kolumbien)**  
den Hochschulgrad

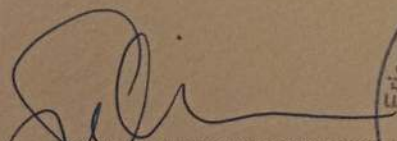
## MASTER OF SCIENCE (M.Sc.)

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Er hat die Masterprüfung (120 ECTS)  
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

  
Prof. Dr. Christian Spielmann  
Dekan



  
Prof. Dr. Silvana Botti  
Vorsitzende des  
Prüfungsausschusses



# Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

## ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

### bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r  
Maria VALLAZZA



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# ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

